EU-ROCKET LAB – BIP 2026

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CELESTIAL HORIZONS

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CONTEXT & CONSTRAINTS

Our EU-ROcket Lab project involves designing an onboard instrumentation solution for an experimental rocket. The aim is to measure data during flight and transmit it after landing via LoRa.

The required hardware is an ESP32 (Heltec LoRa V2) powered by a LiPo battery, with sensors to be selected by the team.

The reference rocket measures approximately 101 cm, with a diameter of 75 mm and a mass of approximately 500 g. It uses an Ø18 mm motor and descends with a 70 cm parachute.

The significant physical constraints to be taken into account are:

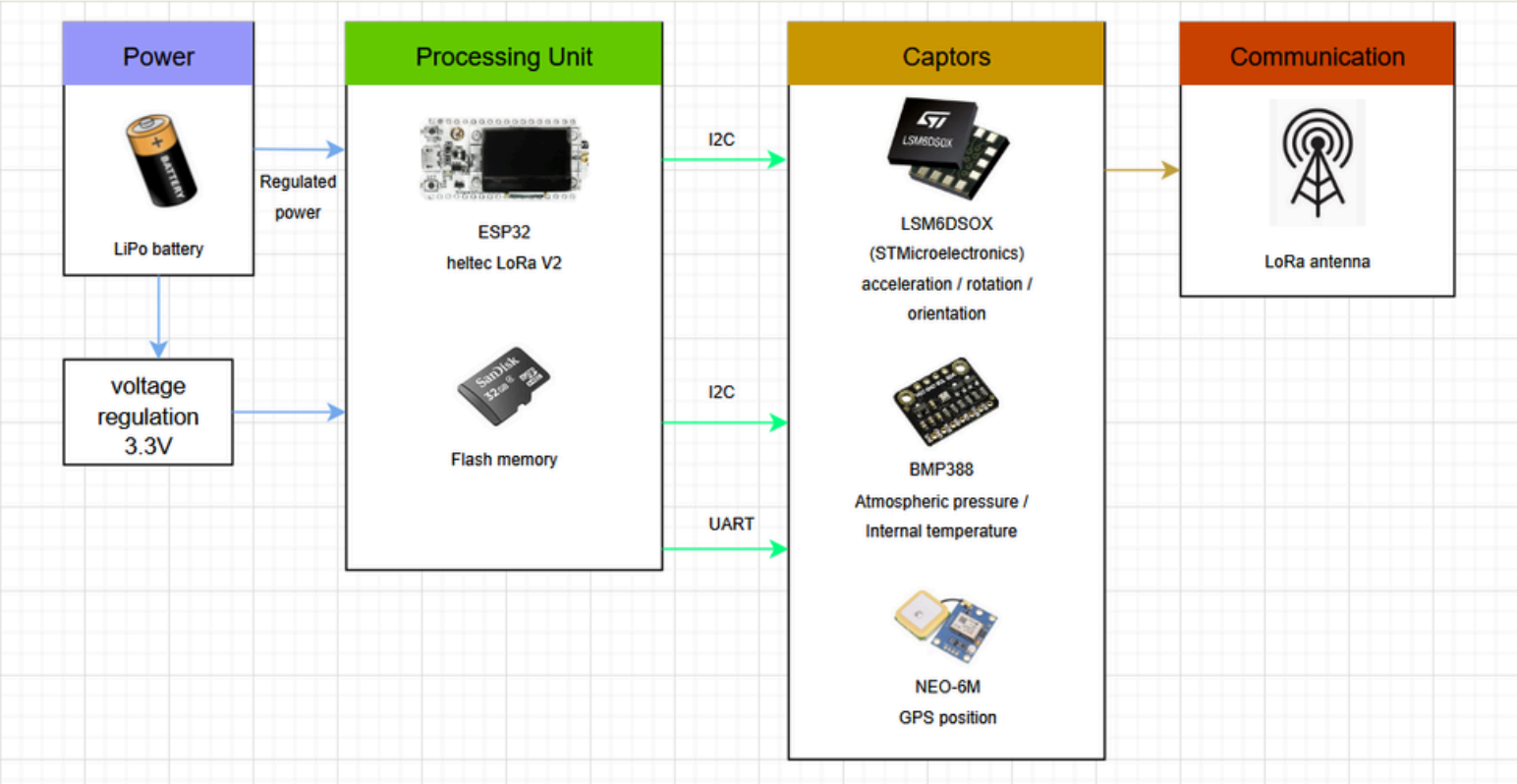
- High acceleration during takeoff
- Significant vibrations during thrust
- Heat transmitted by the motor
- Potential impact on landing
- Limited internal volume for electronics

The system must therefore be lightweight, robust, energy efficient, and capable of operating autonomously throughout the flight.

TECHNOLOGICAL CHOICES

Measurement	Sensor type	Concrete example	Why do we want it?	Sampling phase
Acceleration (X,Y,Z)	High acceleration IMU	LSM6DSOX (STMicroelectronics)	Measure thrust during takeoff, detect flight phases (boost, apogee, descent)	High frequency during launch + flight
Orientation / Rotation	Integrated gyroscope (IMU)	LSM6DSOX	Check flight stability and detect excessive rotation	Throughout the flight
Atmospheric pressure	Barometric sensor	BMP388	Calculate altitude accurately and determine apogee	During ascent + descent
Estimated altitude	Software calculation (via pressure)	—	Analyze flight performance and maximum altitude reached	During ascent + descent
Internal temperature	Built-in thermal sensor	BMP388	Monitor the thermal impact of the engine on the electronics	During flight + after landing
GPS position	UART GPS Module	NEO-6M	Precisely locate the rocket after landing	Mainly after landing

EMBEDDED SYSTEM



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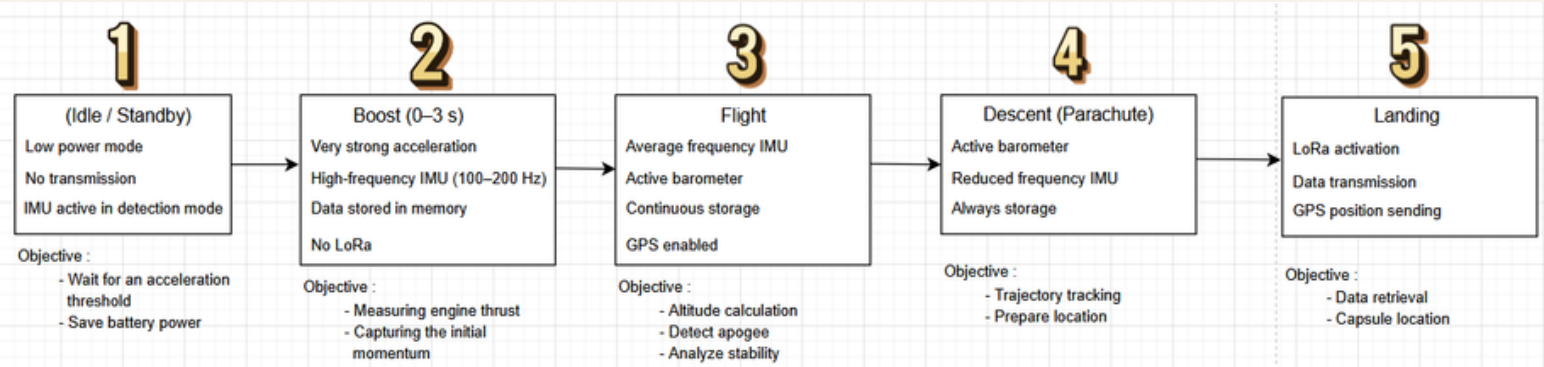

DATA TRANSMISSION

TRANSMISSION SCENARIO:
DURING FLIGHT (MINIMALIST APPROACH):

Simple beacon transmission every 10 seconds including estimated altitude (from BMP388), flight timestamp, and battery status. Acceleration and orientation data are sampled at high frequency but not transmitted in real-time due to LoRa bandwidth limitations.

AFTER LANDING (MAIN STRATEGY):

Sequential LoRa transmission in two phases: First, an immediate summary (first 30 seconds) with maximum altitude, peak acceleration during launch, total flight duration, and GPS coordinates (NEO-6M) for recovery. Second, a complete data dump (2-3 minutes) including all stored acceleration XYZ data, orientation/rotation history, atmospheric pressure profile vs time, and internal temperature during flight.

GROUND STATION:

Hardware architecture consists of a LoRa receiver (ESP32 Heltec LoRa V2 on ground) connected via USB/Serial to a PC. Visualization software includes a serial monitor for raw data, real-time graphs showing altitude and acceleration, and a simple map display with GPS coordinates when available.

IDENTIFIED RISKS AND SOLUTIONS:

Packet Loss: Implement ACK (acknowledgment) with automatic retransmission after 3 failed attempts.
Battery Depletion: Deep sleep mode throughout the entire flight with automatic post-landing activation (detected when acceleration equals zero for 3 seconds).
Large Data Volume: Divide into small packets (≤50 bytes) with CRC checksum and simple data compression.

ENERGY MANAGEMENT

Our system is powered by a LiPo battery, which has limited capacity and must ensure that all components continue to function throughout the mission. The main energy consumers are the ESP32 microcontroller, the onboard sensors (IMU, barometer, GPS), and the LoRa radio module, the latter being particularly energy-intensive during transmission phases.

In order to optimize autonomy, we have adopted an energy management strategy based on the use of the ESP32's low-power modes (light sleep/deep sleep) whenever possible, particularly during the pre-launch phase and between acquisition cycles. The sensors are activated only when necessary, and the sampling frequency is adapted to each phase of the flight (high frequency during boost, lower during descent).

Data is recorded locally in onboard memory during flight and then transmitted via LoRa only after landing. This strategy significantly reduces radio-related consumption while ensuring the integrity and availability of the collected data.

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